URINARY PHTHALATE EXPOSURE AND REPRODUCTIVE OUTCOMES IN U.S. WOMEN: EVIDENCE OF DOSE-RESPONSE ASSOCIATIONS IN NHANES 2003–2014

A THESIS IN
Bioinformatics

Presented to the Faculty of the University
of Missouri-Kansas City in partial fulfillment of
the requirements for the degree

MASTER OF SCIENCE

by
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Bachelor of Pharmacy

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Kansas City, Missouri
2026

Urinary Phthalate Exposure and Reproductive Outcomes in U.S. Women: Evidence of Dose-Response Associations in NHANES 2003–2014

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University of Missouri-Kansas City, 2026

ABSTRACT

Phthalates are a group of endocrine-disrupting chemicals widely used in consumer products and consistently detected in human populations. Although experimental studies suggest that phthalate exposure may affect reproductive function, evidence in women at the population level remains limited and not fully consistent. This study examined the relationship between urinary phthalate metabolites and reproductive outcomes among women aged 18 to 44 years using data from the National Health and Nutrition Examination Survey (NHANES) 2003–2014 (N = 1,806).

Survey-weighted logistic regression models were used to evaluate associations between log-transformed urinary phthalate metabolites and parity, adjusting for age, body mass index, socioeconomic factors, and urinary creatinine. For most metabolites, the estimated associations were close to no meaningful effect. However, mono-benzyl phthalate (MBzP) showed a modest positive association with parity in the fully adjusted model (1.19 (1.02, 1.37)). In quartile-based analyses, higher MBzP exposure levels were associated with increasing odds of parity, with evidence of a linear trend (p-trend = 0.029). These findings were generally consistent across sensitivity analyses, including alternative methods of creatinine adjustment and exclusion of extreme values.

A cumulative exposure index for phthalates did not demonstrate stronger associations than individual metabolites, suggesting that individual compounds may be more informative than the composite exposure summary for this outcome. Secondary analyses of subfertility and menstrual irregularity were more limited due to smaller sample sizes and reduced statistical power.

MBzP showed a modest positive association with parity that warrants cautious interpretation and replication in longitudinal studies using more proximal reproductive endpoints.

APPROVAL PAGE

The faculty listed below, appointed by the Dean of the School of Medicine, have examined this thesis titled "Urinary Phthalate Exposure and Reproductive Outcomes in U.S. Women: Evidence of Dose-Response Associations in NHANES 2003–2014," presented by Anusha Thondepu, candidate for the Master of Science, and certify that in their opinion it is worthy of acceptance.

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ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my advisor, Dr. Sudeepa Bhattacharyya, for her continuous guidance, encouragement, and thoughtful feedback throughout this thesis. Her mentorship has been invaluable to both my academic development and the completion of this work.

I am also grateful to my committee members, Dr. Stephen Simon and Dr. Dua Prerna, for their time, insights, and constructive suggestions. Their feedback played an important role in strengthening this research.

I would like to thank the Department of Biomedical and Health Informatics at the University of Missouri–Kansas City for providing a supportive academic environment and the resources necessary to carry out this work.

Finally, I am especially thankful to my family and friends for their constant encouragement, patience, and unwavering support throughout my graduate journey. Their belief in me helped me stay motivated through every challenge.

CHAPTER 1. INTRODUCTION

Overview of Study

Environmental exposures have become an important area of research in women's reproductive health, as many commonly encountered chemicals may interfere with normal hormonal regulation (Kahn et al., 2020; Gore et al., 2015). Among these, phthalates have received particular attention due to their widespread use in consumer products and their consistent detection in human biomonitoring studies (Hauser & Calafat, 2005; Centers for Disease Control and Prevention, 2021). Because of their potential role as endocrine disruptors, there is growing concern about how these chemicals may influence reproductive function in women (Kahn et al., 2020; Hlisníková et al., 2020).

This thesis examines the relationship between urinary phthalate metabolites and reproductive outcomes among U.S. women aged 18 to 44 years using data from the National Health and Nutrition Examination Survey (NHANES) (Zota et al., 2014). The primary outcome of interest is parity, categorized as nulliparous versus parous. Secondary outcomes include menstrual irregularity and subfertility-related phenotypes, based on the availability of comparable survey data across cycles (Bingru et al., 2025). Survey-weighted regression models were used to account for the complex NHANES sampling design and to estimate associations between phthalate biomarkers and these reproductive outcomes.

This study is important because phthalate exposure is nearly ubiquitous, while reproductive health conditions remain a significant public health concern (Hauser & Calafat, 2005; Swan, 2008). Using nationally representative data allows for evaluation of these relationships at the population level

(Zota et al., 2014). The goal of this thesis is to assess whether urinary phthalate exposure is associated with measurable reproductive phenotypes in women of reproductive age and to contribute to the understanding of environmental influences on reproductive health.

In this study, parity is considered a cumulative reproductive life-history measure rather than a direct indicator of biological fertility. It was selected as the primary outcome because it is the most consistently available reproductive variable across NHANES cycles and allows for a larger analytic sample. In population-based datasets, more direct measures of fertility, such as time to pregnancy or infertility, are often limited to specific cycles or smaller subgroups, reducing statistical power and comparability (Tzouma et al., 2025).

However, parity reflects a combination of biological, behavioral, social, and healthcare-related factors and should not be interpreted as a direct measure of fertility (James-Todd et al., 2016). Accordingly, findings based on parity should be viewed as exploratory and interpreted alongside studies that use more proximal reproductive endpoints.

CHAPTER 2. REVIEW OF LITERATURE

Phthalates are a class of synthetic chemicals widely used in the production of plastics, food packaging materials, personal care products, household items, and certain medical devices (Hauser & Calafat, 2005; Wormuth et al., 2006). These compounds are added to improve flexibility, durability, and stability. However, because phthalates are not chemically bound to the materials in which they are used, they can leach, migrate, or off-gas into the surrounding environment over time (Hauser & Calafat, 2005). This property contributes to continuous and widespread human exposure through multiple pathways, including ingestion, inhalation, and dermal contact (Wormuth et al., 2006). As a result, phthalate metabolites are consistently detected in human urine samples and are commonly used as biomarkers of recent exposure in epidemiologic studies (Barr et al., 2003; Centers for Disease Control and Prevention, 2021).

One of the primary concerns surrounding phthalates is their role as endocrine-disrupting chemicals (Kahn et al., 2020; Gore et al., 2015). Endocrine disruptors can interfere with normal hormonal function, even at relatively low levels of exposure (Gore et al., 2015). Experimental and toxicological studies suggest that certain phthalates may alter steroid hormone synthesis, disrupt ovarian signaling pathways, and contribute to oxidative stress and inflammatory processes (Hannon & Flaws, 2015; Hlisníková et al., 2020). These mechanisms are particularly relevant to female reproductive health, as normal reproductive function depends on tightly regulated hormonal interactions within the hypothalamic–pituitary–ovarian axis (Hannon & Flaws, 2015). Disruptions in this system may affect follicular development, ovulation, menstrual cyclicity, and overall reproductive capacity.

Epidemiologic research has increasingly examined associations between phthalate exposure and reproductive outcomes in women (Swan, 2008; Tzouma et al., 2025). Studies have evaluated outcomes such as infertility, time to pregnancy, menstrual irregularities, hormone concentrations, and assisted reproductive technology (ART) outcomes (Swan, 2008). Some findings suggest that higher phthalate exposure may be associated with altered hormone levels or disruptions in reproductive function; however, results have been inconsistent across studies (Swan, 2008; Tzouma et al., 2025). Differences in study populations, exposure assessment methods, timing of measurement, and outcome definitions contribute to this variability (Tzouma et al., 2025). In addition, many studies rely on cross-sectional data, limiting the ability to establish temporal relationships between exposure and outcome.

An important consideration in this field is the heterogeneity among individual phthalate metabolites (Lyche et al., 2009). Phthalates differ in their chemical structure, sources of exposure, metabolism, and potential biological effects (Lyche et al., 2009). Among these, mono-benzyl phthalate (MBzP) has received particular attention. MBzP is a primary metabolite of butyl benzyl phthalate (BBzP), which has historically been used in materials such as vinyl flooring, adhesives, and sealants (Lyche et al., 2009). This distinguishes MBzP from metabolites more strongly linked to personal care products or dietary sources, suggesting that it may reflect indoor environmental exposures. Evaluating metabolite-specific associations is therefore important for understanding how different exposure pathways may relate to reproductive outcomes.

Phthalate exposures typically occur as mixtures rather than in isolation, and many metabolites are moderately to strongly correlated (Hlisníková et al., 2020). This correlation can complicate statistical modeling and interpretation, particularly when using traditional regression approaches that assess one exposure at a time (Hlisníková et al., 2020). Mixture-based analytical methods, such as weighted quantile sum (WQS) regression and Bayesian kernel machine regression (BKMR), have been developed to evaluate the combined effects of correlated exposures (Carrico et al., 2015; Bobb et al., 2015). These approaches can provide insight into overall exposure burden while identifying influential components within mixtures. However, these methods were not implemented in the present analysis, and the results should therefore be interpreted in the context of single-exposure models and simplified composite measures.

The National Health and Nutrition Examination Survey (NHANES) provides a valuable platform for studying environmental exposures at the population level (Zota et al., 2014). NHANES combines questionnaire data with laboratory-based biomonitoring in a nationally representative sample of the U.S. population (Centers for Disease Control and Prevention, 2021). The availability of urinary phthalate metabolite measurements across multiple survey cycles allows for the evaluation of exposure patterns and their potential associations with reproductive health indicators (Zota et al., 2014).

Despite these strengths, NHANES-based analyses have several important limitations (Tzouma et al., 2025). Phthalate exposure is typically assessed using a single spot urine sample, which may not reflect long-term exposure due to the short biological half-life of these compounds (Barr et al., 2003). Reproductive outcomes are often self-reported, introducing potential recall bias or misclassification (Tzouma et al., 2025). Additionally, the cross-sectional design limits the ability to establish temporality or causal relationships. These limitations should be considered when interpreting findings.

Overall, current evidence suggests ongoing concern regarding phthalates as endocrine-disrupting chemicals with potential implications for women's reproductive health (Kahn et al., 2020; Gore et al., 2015). However, important gaps remain in understanding how individual metabolites relate to specific reproductive phenotypes, particularly in population-based settings (Tzouma et al., 2025). Further research is needed to improve exposure assessment, address mixture complexity, and evaluate these relationships using longitudinal designs. This study contributes to this effort by examining associations between urinary phthalate metabolites and reproductive outcomes using nationally representative data, with a focus on both individual compounds and broader exposure patterns.

Justification

Despite ongoing concern regarding endocrine-disrupting chemicals, important gaps remain in understanding their relationship with women's reproductive health at the population level (Kahn et al., 2020; Gore et al., 2015). Many prior studies have focused on infertility clinic populations or specific reproductive disorders, often with limited exposure assessment (Swan, 2008; Tzouma et al., 2025). In contrast, fewer studies have evaluated parity and related reproductive phenotypes using nationally representative U.S. data with biomonitoring-based exposure measures (Zota et al., 2014). Given the widespread nature of phthalate exposure and the public health importance of reproductive outcomes, further investigation in population-based datasets is warranted (Hauser & Calafat, 2005).

The present study is motivated by the need to better understand whether urinary phthalate metabolites are associated with reproductive outcomes among women of reproductive age in the United States. NHANES provides a valuable platform for this analysis, as it integrates biomonitoring data, standardized reproductive health variables, and a nationally representative sampling design (Centers for Disease Control and Prevention, 2021; Zota et al., 2014). Evaluating these associations at the population level may help clarify the contribution of environmental exposures to reproductive health.

Recent NHANES-based studies have examined associations between endocrine-disrupting chemicals and reproductive outcomes, including infertility, menstrual irregularities, and gynecologic conditions (Bingru et al., 2025; Tzouma et al., 2025). These studies increasingly incorporate mixture-based approaches and highlight the complexity of interpreting correlated exposures (Carrico et al., 2015; Bobb et al., 2015). Despite these advances, findings remain inconsistent, particularly when comparing proximal reproductive outcomes with broader population-level indicators (Tzouma et al., 2025). This underscores the need for cautious interpretation and further longitudinal research.

Overall, prior research suggests that phthalate exposure may be related to reproductive health, but the evidence is mixed across study populations, outcome definitions, and individual metabolites. Many previous studies have used clinical, regional, or selected cohorts, which may limit generalizability, and few studies have evaluated these associations in nationally representative samples of U.S. women across multiple survey cycles. Also, some studies have focused on broad phthalate classes instead of examining whether specific metabolites show different patterns of association. Little is known about whether individual metabolite findings differ from simplified cumulative exposure measures in population-based analyses. The present study was designed to address several of these gaps by using pooled NHANES data, applying appropriate survey-weighted methods, and evaluating both individual urinary phthalate metabolites and summary exposure measures in relation to reproductive phenotypes among women of reproductive age. Based on these considerations, we developed our study questions and hypotheses as stated below.

Study Questions and Hypotheses

This study addresses the following research questions:

1. Are urinary phthalate metabolites associated with parity among U.S. women of reproductive age?
2. Are urinary phthalate metabolites associated with menstrual irregularity and subfertility-related phenotypes?
3. Do individual metabolites show different patterns than cumulative measures?
4. Do selected phthalate metabolites demonstrate dose-response trends across increasing exposure levels?

It is hypothesized that selected phthalate metabolites may be associated with altered reproductive phenotypes among women of reproductive age, and the direction and magnitude of associations may vary across metabolites. It is further hypothesized that some individual metabolites may exhibit more distinct exposure-response patterns than cumulative exposure measures.

Research Objectives

The primary objective of this study is to examine associations between urinary phthalate exposure and parity among U.S. women aged 18 to 44 years using NHANES data.

The secondary objectives are:

1. To evaluate associations between urinary phthalate exposure and menstrual irregularity where relevant questionnaire data are available.
2. To evaluate associations between urinary phthalate exposure and subfertility-related phenotypes, including prolonged time to pregnancy, where data are available.
3. To assess whether dose-response patterns are present across increasing levels of exposure for selected phthalate metabolites.
4. To examine whether individual urinary phthalate metabolites demonstrate stronger associations with reproductive outcomes than cumulative exposure measures.

CHAPTER 3. METHODOLOGY

Study Population

Data for this study were obtained from the National Health and Nutrition Examination Survey (NHANES), a nationally representative survey of the civilian, noninstitutionalized U.S. population conducted by the National Center for Health Statistics. NHANES uses a complex, multistage probability sampling design and combines in-home interviews with standardized physical examinations and laboratory testing conducted in Mobile Examination Centers (MECs). This design enables the collection of both self-reported and objectively measured health data across diverse population groups.

The primary analysis used pooled survey cycles from 2003 to 2014 to increase sample size and improve the stability of estimates. Pooling multiple cycles is particularly important when examining environmental exposures and reproductive outcomes, which may require sufficient variability and sample size for reliable estimation.

The analytic sample was restricted to women aged 18 to 44 years at the time of examination, representing the reproductive-age population. Participants were included if they completed the MEC examination and had available data on urinary phthalate metabolites, reproductive outcomes, and relevant covariates required for the regression models. Individuals with missing data on key exposure, outcome, or covariate variables were excluded from the final analytic sample. The stepwise sample selection process is illustrated in Figure 1.

Study Outcomes

The primary outcome in this study was parity, operationalized as a binary variable distinguishing between nulliparous and parous women. Parity was conceptualized as a cumulative reproductive life-history measure reflecting a combination of biological, behavioral, and social influences over time. Although parity does not directly measure biological fertility, it provides a summary indicator of reproductive experience and has been widely used in epidemiologic research.

This binary categorization was selected to facilitate interpretation and to align with the use of survey-weighted logistic regression models. Treating parity as a binary outcome also reduces the influence of skewed distributions that may arise when parity is modeled as a count variable.

Secondary outcomes included menstrual irregularity and subfertility-related phenotypes. Menstrual irregularity was defined based on self-reported questionnaire responses describing cycle patterns, where consistent definitions were available across survey cycles. Subfertility-related phenotypes were assessed using self-reported measures, including attempts to become pregnant for more than 12 months without success. These outcomes were included to capture additional dimensions of reproductive function beyond cumulative reproductive history.

Smoking was considered as a potential confounder, given its established associations with both environmental exposures and reproductive outcomes, and was accounted for in the interpretation of the findings.

Study Procedures

Urinary phthalate metabolites were the primary exposure variables in this study. The metabolites examined included mono-ethyl phthalate (MEP), mono-n-butyl phthalate (MnBP), mono-isobutyl phthalate (MiBP), mono-benzyl phthalate (MBzP), mono-(3-carboxypropyl) phthalate (MCPP), and selected metabolites of di(2-ethylhexyl) phthalate (DEHP), including MEHP, MEHHP, and MEOHP. These metabolites were selected based on their availability across NHANES cycles and their relevance in prior environmental health research.

Urinary phthalate concentrations were evaluated prior to analysis to assess distributional properties. Because these biomarkers were right-skewed, natural logarithmic transformation was applied to improve distributional symmetry and reduce the influence of extreme values. Values below the limit of detection (LOD) were handled using NHANES-provided imputed values, typically calculated as the LOD divided by the square root of 2. This approach allows for consistent inclusion of low-level exposure values and follows standard NHANES recommendations. Log transformation was applied after incorporating these values.

All phthalate metabolite concentrations were analyzed in their original measurement units prior to transformation, and urinary creatinine was included as a covariate in all regression models to account for variation in urine dilution. In addition to continuous exposure modeling, exposure variables were categorized into quartiles for selected analyses to assess potential dose–response relationships and to facilitate interpretability across exposure levels.

Potential confounders were selected a priori based on prior literature and biological plausibility. These covariates included age (RIDAGEYR), body mass index (BMXBMI), education level (DMDEDUC2), poverty-income ratio (INDFMPIR), and race/ethnicity (RIDRETH1). These factors are associated with both environmental exposures and reproductive health outcomes and were included to reduce confounding.

To ensure consistency across pooled survey cycles, variables were harmonized where necessary. In particular, race and ethnicity categories were aligned across cycles to maintain comparability in the analytic dataset. All variable definitions and coding decisions were reviewed to ensure consistency and reproducibility.

Missing Data

Missing data were handled using a complete-case approach for each model. Participants with missing information on the exposure, outcome, or covariates relevant to a given analysis were excluded from that specific model. As a result, the analytic sample size varied across primary, secondary, and sensitivity analyses.

Although complete-case analysis may reduce sample size, it allows for consistent model estimation using observed data without introducing additional assumptions required for imputation. Given the cross-sectional design of NHANES and the complexity of survey weighting, this approach was considered appropriate for the primary analyses. The potential impact of missing data on study findings was considered during interpretation, particularly for secondary outcomes with smaller analytic samples.

Statistical Analysis

All analyses accounted for the complex NHANES sampling design by incorporating survey weights, stratification, and primary sampling units. Because urinary phthalate metabolites were measured in subsamples and the corresponding weight variables differed across survey cycles, the appropriate subsample weights were identified separately for each cycle (e.g., WTSPH2YR for earlier cycles and WTSB2YR for later cycles). These cycle-specific weights were harmonized to create a pooled phthalate weight variable. To construct pooled weights across the 2003–2014 survey period, each 2-year subsample weight was divided by the number of combined cycles (n = 6), following NHANES analytic guidelines. Survey design variables, including strata (SDMVSTRA) and primary sampling units (SDMVPSU), were specified in all analyses to ensure nationally representative estimates.

Survey-weighted logistic regression models were used to estimate associations between urinary phthalate metabolites and binary reproductive outcomes. The primary analysis focused on parity (nulliparous vs. parous). Secondary analyses examined menstrual irregularity and subfertility-related phenotypes where sufficient data were available.

All models were adjusted for a set of a priori selected covariates, including age, body mass index, education level, poverty-income ratio, race/ethnicity, and urinary creatinine. These variables were selected based on prior literature and their potential to confound associations between environmental exposures and reproductive outcomes. Results were reported as odds ratios (ORs) with corresponding 95% confidence intervals (CIs).

NHANES cycles were combined by harmonizing variable definitions across survey years and appending participant-level data into a single pooled dataset after applying consistent inclusion criteria. Care was taken to ensure consistent coding across cycles, particularly for demographic and reproductive variables.

All statistical analyses were conducted in R using the *survey* package to account for the complex sampling design. Data management and variable transformation were performed using the *dplyr* package, supporting reproducible analysis workflows.

To evaluate potential dose–response relationships, selected phthalate metabolites were modeled using categorized exposure variables. Exposure distributions were divided into quartiles based on the distribution of log-transformed concentrations within the analytic sample. These categories were included in regression models as indicator variables, with the lowest exposure group serving as the reference category. Linear trends across quartiles were assessed by modeling the quartile variable as an ordinal term.

To account for multiple comparisons in the primary parity analyses, false discovery rate (FDR) adjustment was applied using the Benjamini–Hochberg procedure across metabolite-specific tests. The cumulative phthalate index was evaluated separately and was not included in the FDR-adjusted set, as it represents a distinct exposure construct.

Sensitivity analyses were conducted to assess the robustness of the findings. These included alternative exposure specifications (e.g., creatinine-corrected models), exclusion of extreme exposure values, and stratified analyses to evaluate potential heterogeneity across subgroups.

Finally, effect modification by age was evaluated by including an interaction term between MBzP and age group (18–24, 25–34, and 35–44 years) in a survey-weighted logistic regression model.

CHAPTER 4. RESULTS

4.1 Analytic Sample Characteristics

The final study population consisted of 1,806 women aged 18 to 44 years drawn from pooled NHANES cycles 2003 to 2014 after applying eligibility criteria and excluding records with incomplete data required for the primary adjusted analyses. The stepwise derivation of the analytic sample is summarized in Figure 1, and descriptive characteristics are presented in Table 1.

The reduction from the eligible sample to the final analytic sample was primarily driven by missing covariate data, including body mass index, poverty-income ratio, and education.

Within the survey-weighted analytic population, the mean age was 32.18 years (SE 0.25), and 66.9% of participants were classified as parous (95% CI: 63.2–70.3). The weighted racial and ethnic composition was 62.8% non-Hispanic White (95% CI: 59.2–66.4), 12.6% non-Hispanic Black (95% CI: 10.6–14.5), 9.9% Mexican American (95% CI: 8.0–11.9), 6.8% other Hispanic (95% CI: 5.3–8.3), and 7.9% other race (95% CI: 6.4–9.4). The weighted mean body mass index was 28.32 kg/m^2 (SE 0.20), and the weighted mean poverty-income ratio was 2.74 (SE 0.06).

Urinary phthalate biomarkers exhibited positively skewed distributions; therefore, logarithmic transformation was applied prior to model fitting. This preprocessing step improved comparability across metabolites and enhanced the stability of regression estimates.

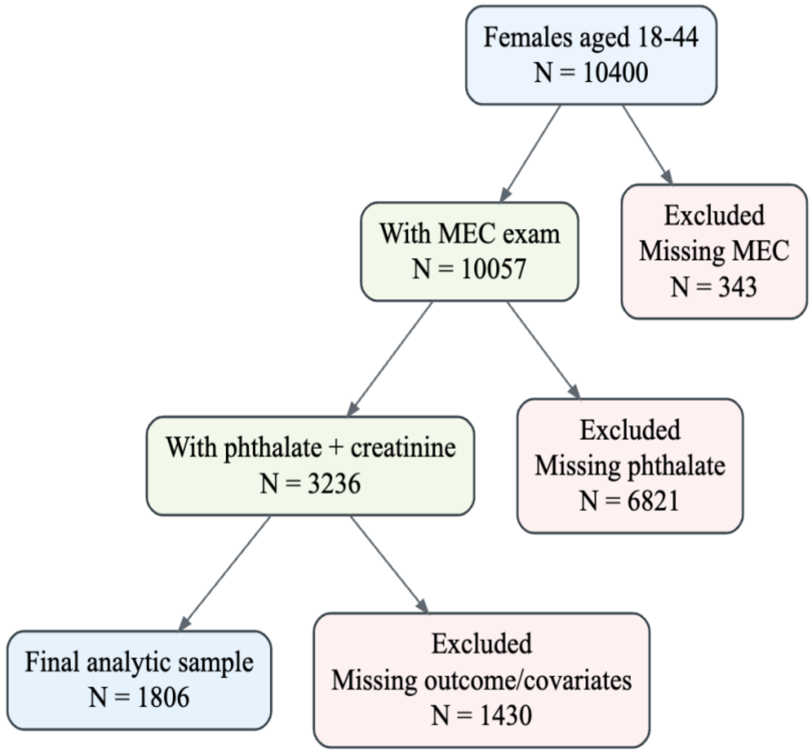

Figure 1* *Flowchart of analytic sample selection (N = 1,806).*

4.2 Primary Associations Between Phthalates and Parity

4.2.1 Individual Metabolite Models

The primary regression analyses indicated a modest association for MBzP relative to other phthalate metabolites (Table 2; Figure 2). In the fully adjusted survey-weighted model, higher MBzP levels were associated with slightly higher odds of parity (OR 1.19; 95% CI: 1.02–1.37; p-value = 0.025). However, this association was attenuated after adjustment for multiple comparisons and should be interpreted with caution.

The remaining metabolites showed estimates closer to the null. MnBP (OR 1.04; 95% CI: 0.88–1.24), MiBP (OR 1.05; 95% CI: 0.91–1.22), and MCPP (OR 1.11; 95% CI: 0.67–1.83) did not demonstrate clear patterns of association. MCPP showed the widest confidence interval, indicating lower precision relative to other metabolites.

A similar pattern was observed among the DEHP metabolites. MEHP, MEHHP, and MEOHP showed point estimates above 1.0 and moved in the same direction; however, the magnitude of these associations remained modest and confidence intervals included the null value (figure-6).

Taken together, the individual metabolite models did not demonstrate consistent or strong associations across most phthalates. Although MBzP showed a positive association in the fully adjusted model, this finding was not statistically robust after correction for multiple comparisons and should be interpreted cautiously.

Adjustment for demographic, socioeconomic, and urinary dilution variables reduced the MBzP effect estimate relative to simpler models, suggesting that part of the crude association may be explained by covariate structure. Although the direction of association remained positive after adjustment, the magnitude was modest.

Given that parity reflects cumulative reproductive history rather than a direct measure of biological fertility, these findings should be interpreted as associations with population-level reproductive patterns rather than evidence of effects on fertility.

An additional consideration in interpreting the DEHP-related findings is the strong correlation among individual DEHP metabolites. MEHP, MEHHP, and MEOHP are derived from the same parent compound and reflect related metabolic pathways following exposure. Because of this shared source, these biomarkers tend to move together rather than represent fully independent exposures. In epidemiologic analyses, such correlation may result in similar directional patterns but reduced statistical precision for individual estimates, which may contribute to the modest and non-significant findings observed for this group.

4.2.2 Cumulative Exposure Index

To assess broader combined exposure patterns, an exploratory composite exposure index was constructed by summing z-scored log-transformed metabolites. In the fully adjusted survey-weighted model, a one standard deviation increase in this index was associated with slightly higher odds of parity (OR 1.04; 95% CI: 0.996–1.09; p-value = 0.070) (Table 2; Figure 2). A similar direction was observed for ΣDEHP, which also showed a modest positive point estimate.

Although both summary measures showed positive point estimates, confidence intervals included the null value, indicating limited statistical precision. These findings do not suggest strong evidence of association at the level of the composite exposure measures.

Compared with individual metabolite models, the composite measures did not demonstrate stronger or more consistent associations.

This composite index assumes equal contribution of each metabolite and does not account for differences in biological potency or potential interactions between exposures. Therefore, it should be interpreted as an exploratory summary measure rather than a biologically weighted exposure metric.

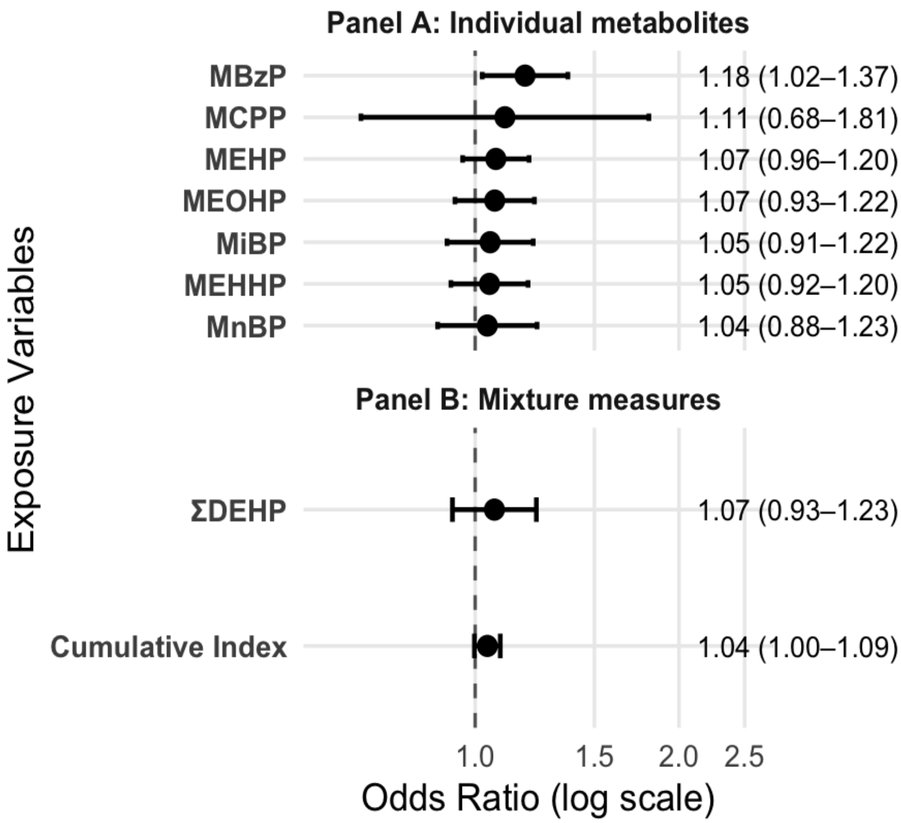

Figure 2* *Adjusted associations between urinary phthalates and parity (N = 1,806). Odds ratios (95% CI) shown on a log scale. Panel A: individual metabolites; Panel B: mixture measures.*

4.3 Secondary Reproductive Outcomes

4.3.1 Subfertility

The weighted prevalence of subfertility in the eligible subgroup was 69.0%. This estimate reflects the restricted eligible subgroup defined by the questionnaire pathway and should not be interpreted as population prevalence of infertility. Across subfertility analyses, estimated associations varied across exposures and model specifications, with generally wide confidence intervals indicating limited precision. MBzP showed a direction similar to that observed in the parity models; however, the estimates were not statistically robust and were accompanied by substantial uncertainty. Overall, no consistent pattern of association was observed for subfertility.

These analyses were limited by smaller sample size and reduced statistical power compared to the primary parity models. As a more proximal reproductive outcome, subfertility may be less influenced by social and behavioral factors than parity; however, the limited analytic sample restricts the ability to draw firm conclusions.

4.3.2 Menstrual Irregularity *and Amenorrhea (2013–2014 Only)*

Menstrual irregularity and amenorrhea were available only in the 2013–2014 survey cycle, resulting in a substantially smaller analytic sample. Within this restricted dataset, MiBP showed associations in selected model specifications; however, estimates varied across models and were accompanied by wide confidence intervals, indicating considerable uncertainty.

Because these outcomes were limited to a single cycle and a reduced analytic sample, findings should be interpreted as exploratory and not as definitive evidence of association.

4.4 Model Stability and Diagnostics

Sample size remained consistent across the primary metabolite models, with 1,806 observations contributing to most analyses. This consistency indicates that differences in effect estimates and precision across metabolites were not primarily driven by variation in analytic sample size.

The correlation structure of urinary phthalate metabolites demonstrated clustering rather than uniform relationships across exposures (Figure 3). The oxidative DEHP metabolites were highly correlated, with pairwise correlations ranging from 0.83 to 0.98. MBzP showed moderate correlations with MnBP ($r = 0.73$) and MiBP ($r = 0.61$), and weaker correlations with DEHP metabolites ($r = 0.44$ to 0.58).

These correlation patterns have implications for model interpretation. High correlation among DEHP metabolites suggests that including multiple related exposures in a single model may reduce precision and complicate interpretation of individual coefficients. In contrast, the moderate correlation profile of MBzP indicates that its observed association is less likely to reflect the shared structure of the DEHP metabolites.

Overall, individual metabolite models did not demonstrate strong or consistent associations across most phthalates. The observed MBzP finding should therefore be interpreted as exploratory and hypothesis-generating rather than definitive.

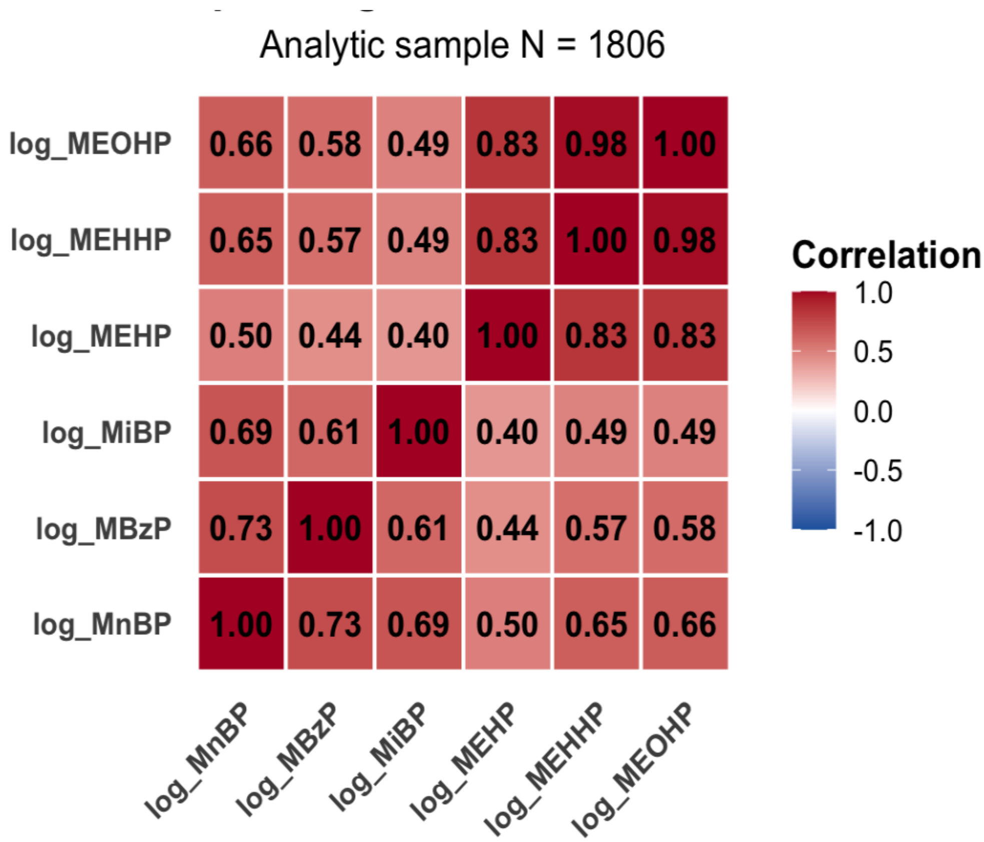

Figure 3* *Correlation heatmap of log-transformed urinary phthalate metabolites among women in the main analytic sample (N = 1,806).*

4.5 Sensitivity Analyses

The MBzP estimate remained generally consistent across a range of alternative model specifications (Table 3; Figure 4). When urinary dilution was handled using a creatinine-corrected exposure approach, the effect estimate was similar in magnitude (OR 1.21; 95% CI: 1.04–1.41), indicating that results were not sensitive to the method of dilution adjustment.

Age-stratified analyses showed similar directions of association across reproductive-age groups. Confidence intervals were wider within strata due to smaller subgroup sizes, indicating reduced precision. Exclusion of the highest 1% of exposure values produced a similar estimate (OR 1.18; 95% CI: 1.01–1.37), suggesting that results were not driven by extreme observations.

Overall, sensitivity analyses demonstrated that the direction of the MBzP association was consistent across alternative specifications, although estimates remained modest in magnitude.

An interaction analysis was conducted to assess whether the association between MBzP and parity differed across age groups. The overall MBzP-by-age-group interaction was statistically significant (p-interaction = 0.027), indicating potential variation across reproductive-age strata. In the interaction model, the interaction term was not statistically significant for ages 25–34 years but was statistically significant for ages 35–44 years, suggesting a weaker association in the oldest age group relative to the reference category.

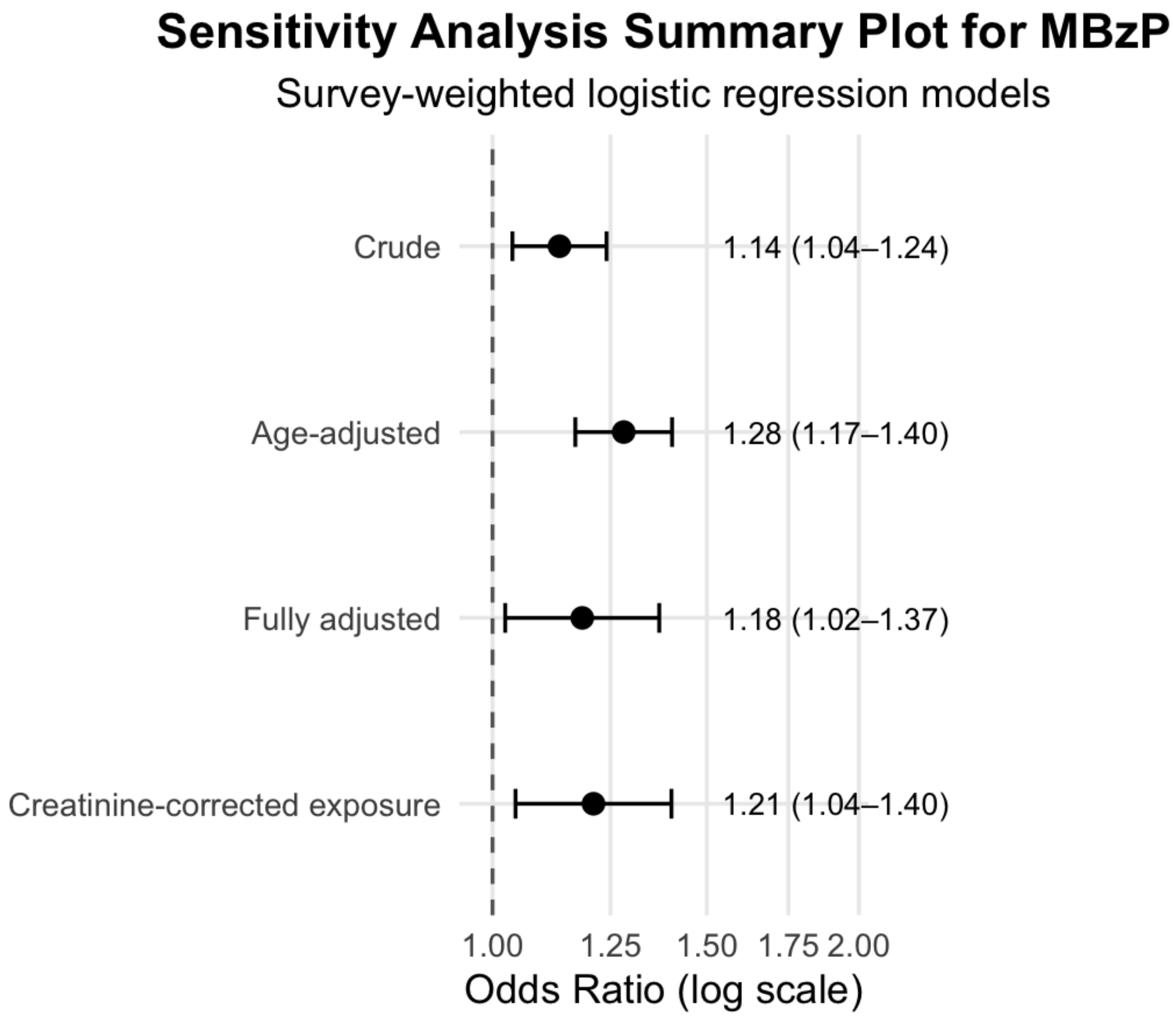

Figure 4* *Sensitivity analysis of the association between MBzP and parity (N = 1,806). Estimates from multiple model specifications are presented for comparison across sensitivity analyses.*

4.6 Dose–Response Analysis

To further examine exposure–gradient patterns, MBzP was evaluated using categorized exposure levels. Quartile-based estimates are presented in Table 4, and both MBzP and cumulative index results are visualized in Figure 6. Relative to the lowest exposure quartile, higher quartiles of MBzP were generally associated with higher odds of parity. The second quartile showed little departure from the reference category (OR 0.91; 95% CI: 0.61–1.38), whereas the third (OR 1.42; 95% CI: 0.82–2.47) and fourth quartiles (OR 1.65; 95% CI: 0.96–2.85) showed larger point estimates. Although point estimates increased across higher exposure categories, confidence intervals were wide and included the null value, indicating limited precision.

When modeled ordinally, the quartile-based trend was statistically significant (p-trend = 0.029). However, given the imprecision of individual category estimates, this pattern should be interpreted cautiously.

To assess the robustness of these patterns, exposure was categorized into quartiles. Quartile-based analyses showed a similar directional pattern, with higher exposure categories corresponding to modestly higher odds of parity; however, confidence intervals remained wide and included the null value, indicating limited precision.

The quartile analysis of the composite exposure summary showed a similar upward pattern, with higher categories corresponding to higher point estimates and a statistically significant trend (p-trend = 0.006). However, confidence intervals overlapped the null value across intermediate categories, indicating imprecision. This composite measure reflects overall exposure patterns and does not account for differences in biological potency or interactions among metabolites; therefore, it should be interpreted as an exploratory summary measure.

The descriptive distribution of MBzP by parity status provided a visual complement to the regression findings. Although exposure distributions overlapped substantially between groups, the distribution among parous women appeared modestly shifted toward higher values relative to nulliparous women.

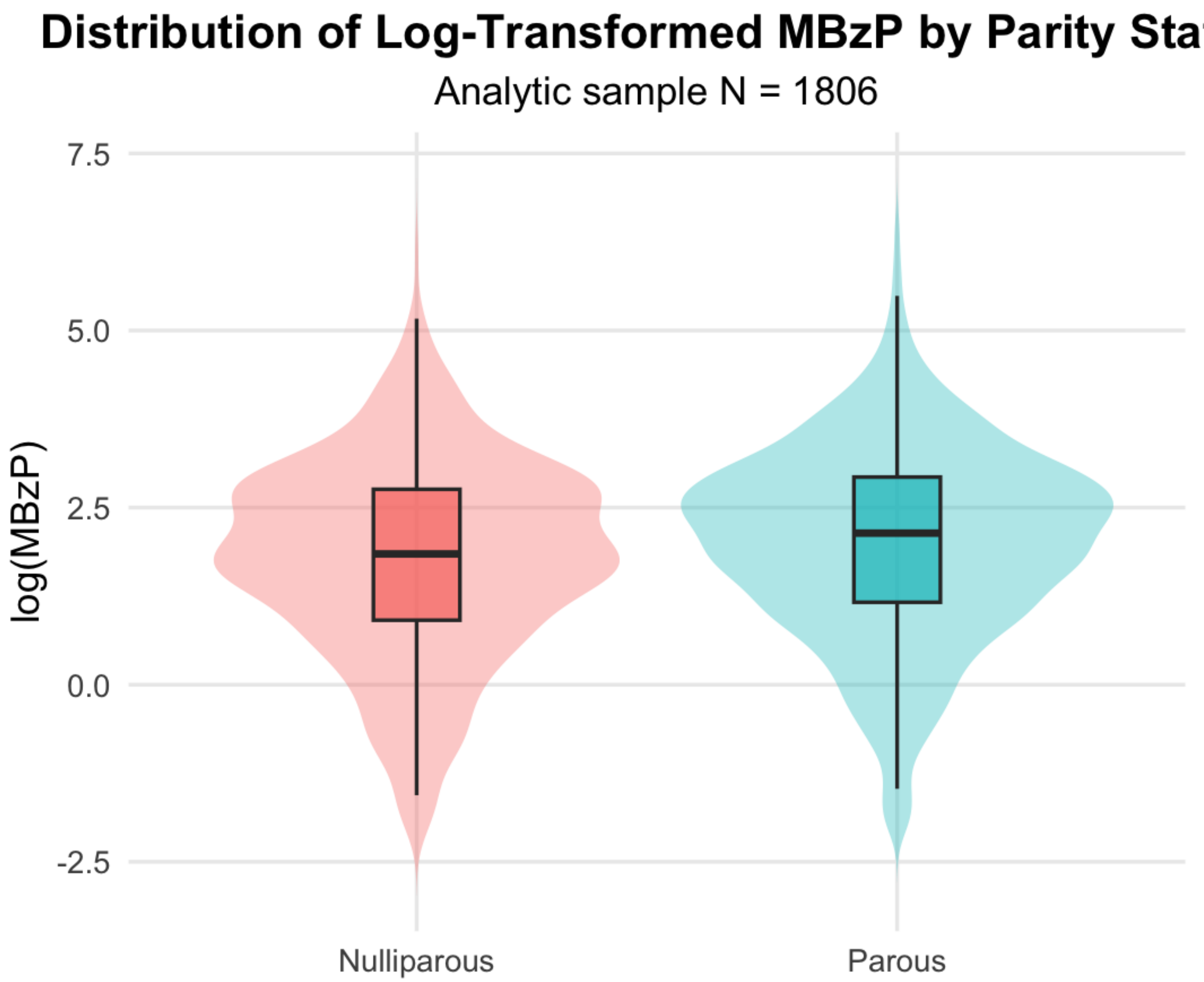

Figure 5* *Distribution of log-transformed MBzP by parity status (N = 1,806). Violin and boxplots are shown for nulliparous and parous groups.*

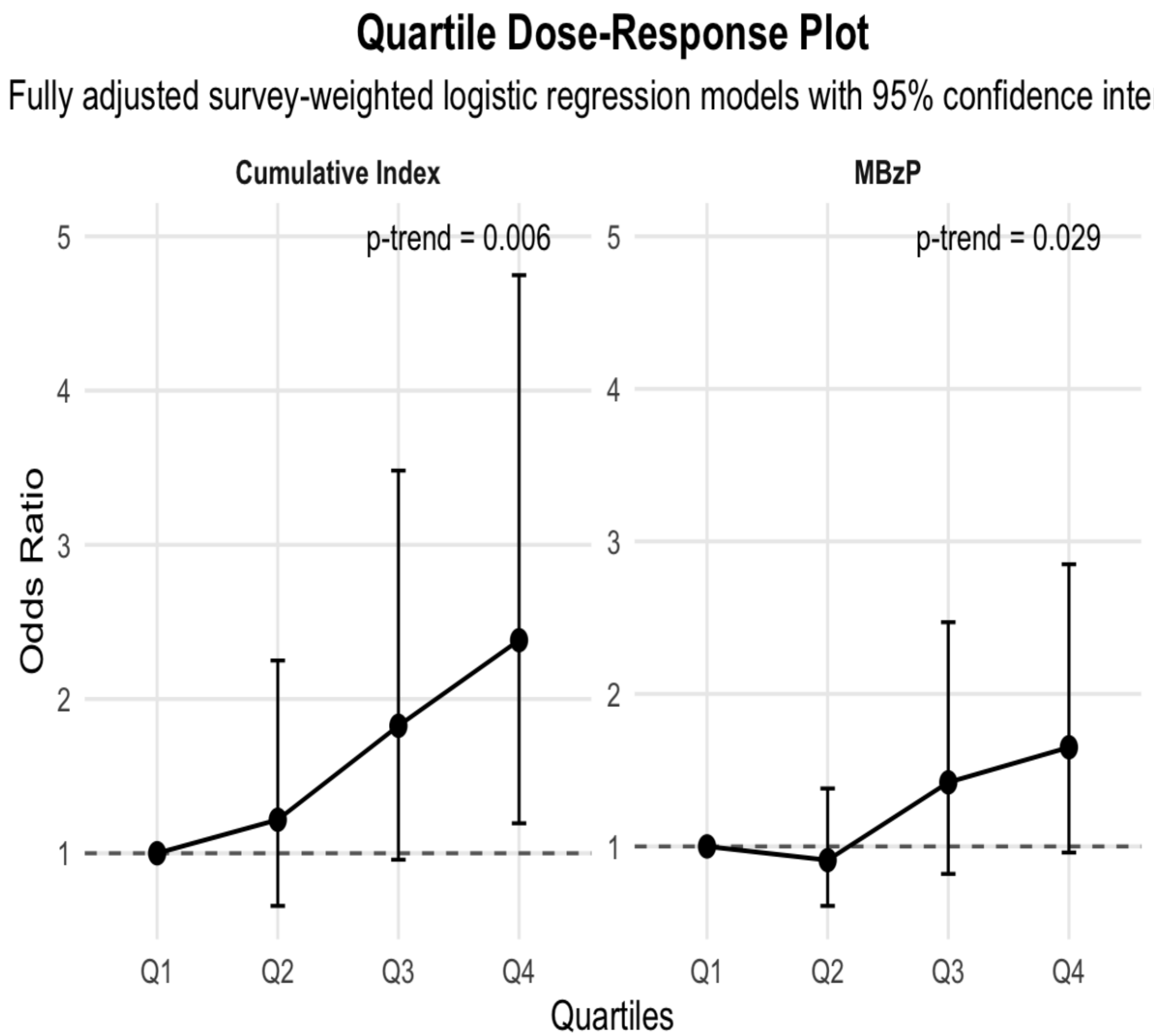

Figure 6* *Quartile analysis of MBzP and composite exposure summary in relation to parity (N = 1,806). Adjusted odds ratios (95% CI) across exposure quartiles for MBzP and composite exposure index (N= 1,806). Confidence intervals are shown for each category.*

Table 1* *Characteristics of the analytic sample (N = 1,806)*

Variable	Mean(SE)/%
Age(years)	32.18 (0.25)
Body Mass Index (kg/m^2)	28.32 (0.20)
Poverty-Income Ratio (PIR)	2.74 (0.06)
Parity (%)	66.9% (95% CI: 63.2–70.3)
Race/Ethnicity	
Non-Hispanic White	62.8% (95% CI: 59.2–66.4)
Non-Hispanic Black	12.6% (95% CI: 10.6–14.5)
Mexican American	9.9% (95% CI: 8.0–11.9)
Other Hispanic	6.8% (95% CI: 5.3–8.3)
Other race	7.9% (95% CI: 6.4–9.4)
Education	
Less than 9th grade	3.2%
9th to 11th grade	10.6%
High School graduate or GED	20.5%
Some college or AA degree	36.3%
College graduate or above	29.5%
Median log-transformed urinary phthalate metabolites	
log(MnBP)	3.043
log(MBzP)	2.181
log(MCPP)	-0.852
log(MiBP)	1.825
log(MEHP)	0.788
log(MEHHP)	2.879
log(MEOHP)	2.480

Table 2* *Association between phthalate metabolites and parity in fully adjusted survey-weighted models*

Exposure	N	OR (95% CI)	p-value	FDR-adjusted p-value
log_MnBP	1,806	1.04 (0.88-1.24)	0.633	0.687
log_MBzP	1,806	1.19 (1.02-1.37)	0.025	0.201
log_MCPP	1,806	1.11 (0.67-1.83)	0.687	0.687
log_MiBP	1,806	1.05 (0.91-1.22)	0.498	0.664
log_MEHP	1,806	1.07 (0.96-1.22)	0.226	0.602
log_MEHHP	1,806	1.05 (0.92-1.20)	0.469	0.664
log_MEOHP	1,806	1.07 (0.93-1.23)	0.338	0.664
log_sumDEHP	1,229	1.11 (0.96-1.29)	0.170	0.602
Cumulative Index (Z-score)	1,229	1.04 (0.996-1.09)	0.070	_

Table 3* *Sensitivity analyses for MBzP and parity*

Sensitivity Type	**OR**	**95% CI**	**Interpretation**
Primary model	1.185	1.022-1.374	Baseline fully adjusted model
Creatinine-corrected exposure	1.210	1.042-1.406	No meaningful change; results robust to dilution handling
Age 18-29	1.171	0.946-1.448	Similar effect size but not significant; likely reduced power
Age 30-44	1.172	0.968-1.419	Consistent with primary model; no strong age modifications
Exclude Top 1%	1.178	1.013-1.370	No meaningful change; not driven by extreme values

Table 4* *Quartile analysis of MBzP and parity*

Quartile	OR	95% CI
Q1 (ref)	1.00	-
Q2	0.91	0.61, 1.38
Q3	1.42	0.82, 2.47
Q4	1.65	0.96, 2.85

Reference group = Q1 (lowest MBzP quartile).

CHAPTER 5: DISCUSSION

This study examined associations between urinary phthalate metabolites and reproductive outcomes among U.S. women of reproductive age using NHANES data. The primary analysis focused on parity, modeled as nulliparous versus parous, while secondary analyses considered subfertility and menstrual irregularity where data were available. Overall, most urinary phthalate metabolites were not strongly associated with parity after adjustment for demographic, socioeconomic, and biologic covariates. MBzP showed a modest positive association and a suggestive dose–response pattern; however, this finding was not statistically robust after adjustment for multiple comparisons and should be interpreted with caution.

A key issue in interpreting these findings is the use of parity as an outcome. In this study, parity is considered a cumulative reproductive life-history measure rather than a direct indicator of biological fertility. It reflects a combination of biological, behavioral, and social factors, including reproductive intentions, contraceptive use, and access to healthcare (James-Todd et al., 2016). Accordingly, the present analysis evaluates associations with population-level reproductive patterns rather than direct effects on biological fertility, and findings should be interpreted as exploratory.

The observed MBzP association requires careful interpretation. Because urinary phthalate metabolites reflect recent exposure whereas parity reflects cumulative reproductive history, reverse causation is a primary concern (Barr et al., 2003). Differences in exposure levels may reflect variation in household environments, product use, or life-stage factors associated with having children rather than a direct biological effect (Wormuth et al., 2006). MBzP, as a metabolite of butyl benzyl phthalate (BBzP), may capture exposure patterns related to indoor environmental materials such as vinyl flooring and adhesives (Lyche et al., 2009). Although experimental studies suggest that phthalates may influence endocrine signaling and ovarian function (Hannon & Flaws, 2015; Hlisníková et al., 2020), the present cross-sectional analysis cannot establish causal relationships or evaluate underlying biological mechanisms.

The broader pattern of results is also informative. Most metabolites showed estimates close to the null, and the cumulative phthalate index did not demonstrate stronger associations than individual metabolites. Secondary analyses of subfertility and menstrual irregularity were limited by smaller analytic samples and reduced statistical precision, and did not show consistent patterns of association. These findings suggest that broad conclusions regarding phthalates and reproductive outcomes are not supported by this dataset. Instead, results indicate that individual metabolites may exhibit distinct, but modest and uncertain, associations under specific conditions, consistent with prior epidemiologic variability (Swan, 2008; Tzouma et al., 2025).

The strong correlation observed among DEHP metabolites is an additional consideration. Because these biomarkers are derived from a common parent compound, they tend to move together and do not represent independent exposures (Hlisníková et al., 2020). In this context, correlated exposures may show similar directional patterns while limiting the ability to detect distinct associations for individual metabolites. This may explain the modest and non-significant findings observed for DEHP-related measures and the lack of stronger associations for composite exposure indices.

Residual confounding is also possible. Smoking, in particular, may be associated with both environmental exposure patterns and reproductive history (Hauser & Calafat, 2005). Although considered in interpretation, it could not be fully accounted for and may influence observed associations.

The age-interaction analysis suggested that the association between MBzP and parity may vary across reproductive-age groups, with a weaker association observed among women aged 35–44 years. However, this finding should be interpreted cautiously, as parity is strongly age-dependent and the interaction analysis was exploratory.

Overall, the cross-sectional design of NHANES limits causal inference (Zota et al., 2014). Because exposure and outcome are measured at the same time point, the temporal relationship between phthalate exposure and reproductive history cannot be determined. As a result, the observed associations should be interpreted as non-causal and hypothesis-generating.

Limitations

Several limitations should be considered when interpreting these findings. First, the cross-sectional design of NHANES limits causal inference and does not allow for clear assessment of temporal relationships between exposure and reproductive outcomes (Zota et al., 2014). Because exposure and outcome are measured at the same time point, it is not possible to determine whether phthalate exposure preceded or followed the observed reproductive patterns.

Second, urinary phthalate metabolites have relatively short biologic half-lives and primarily reflect recent exposure (Barr et al., 2003). As a result, a single spot urine sample may not adequately capture longer-term exposure patterns relevant to cumulative reproductive outcomes such as parity. This limitation may introduce exposure misclassification and could attenuate observed associations.

Third, the interpretation of parity requires caution. Parity reflects a combination of biological, behavioral, and social factors, including reproductive intentions, contraceptive use, partnership status, and access to healthcare (James-Todd et al., 2016). It should therefore not be interpreted as a direct measure of biological fertility, but rather as a broader reproductive life-history indicator.

Fourth, some secondary reproductive outcomes, including subfertility and menstrual irregularity, were available only in specific NHANES cycles. This resulted in smaller analytic samples and reduced statistical power, limiting the ability to detect modest associations and increasing uncertainty in these estimates.

Fifth, reproductive outcomes in NHANES are based on self-reported questionnaire data and may be subject to recall error or misclassification (Tzouma et al., 2025). In addition, residual confounding from unmeasured or imperfectly measured factors, such as lifestyle behaviors or environmental co-exposures, cannot be fully ruled out.

Finally, phthalate exposures occur as complex mixtures (Hlisníková et al., 2020). Although this study examined both individual metabolites and a cumulative exposure index, the analytic approach does not fully capture potential interactions or differences in biological potency among correlated exposures. More advanced mixture modeling approaches, such as weighted quantile sum regression and Bayesian kernel machine regression, may provide additional insight (Carrico et al., 2015; Bobb et al., 2015). In addition, exploratory interaction analyses suggested potential variation in associations across age groups, which should be further evaluated in longitudinal studies using more proximal reproductive outcomes and alternative measures such as gravidity.

Future Research

Future research should prioritize longitudinal study designs incorporating repeated measures of phthalate exposure over time. Prospective approaches would provide a clearer understanding of temporality and help address concerns related to reverse causation (Zota et al., 2014). Such designs are particularly important for evaluating reproductive outcomes that develop over longer periods.

Additional work is needed to better understand why specific metabolites, such as MBzP, may show different associations compared to other phthalates. This may reflect differences in exposure sources, including indoor environmental materials, or variation in underlying biological mechanisms (Lyche et al., 2009). Further investigation is needed to clarify whether observed associations are driven by exposure patterns, biologic effects, or both.

Future studies should also incorporate mixture-based analytical methods and more detailed clinical measures of reproductive function. Approaches such as weighted quantile sum regression or Bayesian kernel machine regression may better capture the combined effects of correlated exposures (Carrico et al., 2015; Bobb et al., 2015). In addition, the use of alternative reproductive indicators, such as gravidity, may provide a more proximal measure of reproductive capacity and help distinguish biological effects from social and behavioral influences reflected in parity.

Conclusion

In this nationally representative sample of U.S. women, most urinary phthalate metabolites were not strongly associated with parity after adjustment for relevant covariates. MBzP showed a modest positive association and a suggestive dose–response pattern; however, this finding was not statistically robust after adjustment for multiple comparisons and should be interpreted cautiously.

Parity in this study is best understood as a cumulative reproductive life-history measure rather than a direct indicator of biological fertility (James-Todd et al., 2016). As such, the observed associations reflect patterns in reproductive history at the population level and should not be interpreted as evidence of causal effects on fertility.

Overall, these findings suggest that phthalates do not exhibit uniform relationships with reproductive outcomes. Instead, individual metabolites may show distinct but modest and uncertain patterns of association, consistent with prior epidemiologic variability (Swan, 2008; Tzouma et al., 2025). Continued research using longitudinal designs, improved exposure assessment, and more proximal reproductive measures will be important for clarifying these relationships.

APPENDIX

Appendix A. NHANES Variables Used in the Analysis

This appendix summarizes the NHANES variables used in the present study, including urinary phthalate metabolites, reproductive outcome variables, covariates, and survey design variables across the included survey cycles.

Table 5 NHANES variables used in the analysis*

Category	Variable Name	NHANES Code	Description
Exposure	Mono-ethyl phthalate (MEP)	URXMEP	Biomarker of diethyl phthalate exposure
Exposure	Mono-n-butyl phthalate (MnBP)	URXMBP	Biomarker of dibutyl phthalate exposure
Exposure	Mono-isobutyl phthalate (MiBP)	URXMIB	Biomarker of di-isobutyl phthalate exposure
Exposure	Mono-benzyl phthalate (MBzP)	URXMZP	Biomarker of butyl benzyl phthalate exposure
Exposure	Mono-(3-carboxypropyl) phthalate (MCPP)	URXMCP	Non-specific phthalate metabolite
Exposure	Mono-(2-ethylhexyl) phthalate (MEHP)	URXMHP	Primary DEHP metabolite
Exposure	Mono-(2-ethyl-5-hydroxyhexyl) phthalate (MEHHP)	URXMHH	Oxidative DEHP metabolite
Exposure	Mono-(2-ethyl-5-oxohexyl) phthalate (MEOHP)	URXMOH	Oxidative DEHP metabolite
Outcome	Parity (primary outcome)	RHQ171	Number of live births
Outcome	Ever pregnant	RHQ131	Includes pregnancies not resulting in live birth
Outcome	Time to pregnancy	RHQ305	Used to define subfertility
Covariate	Age	RIDAGEYR	Age in years
Covariate	Body Mass Index	BMXBMI	BMI (kg/m^2)
Covariate	Education	DMDEDUC2	Educational attainment
Covariate	Poverty-Income Ratio	INDFMPIR	Socioeconomic status

Category	Variable Name	NHANES Code	Description
Covariate	Race/Ethnicity	RIDRETH1	Self-reported race/ethnicity
Covariate	Urinary Creatinine	URXUCR	Adjustment for urine dilution
Survey Design	Sampling weights	WTSB2YR / WTSPH2YR	Subsample weights
Survey Design	Primary sampling unit	SDMVPSU	Cluster variable
Survey Design	Stratification	SDMVSTRA	Survey strata

APPENDIX B. OUTCOME DEFINITIONS AND CODING

This appendix describes the operational definitions and coding procedures used for the primary and secondary reproductive outcomes.

Parity (Primary Outcome)

Parity was defined using RHQ171 (number of live births):

- Nulliparous: 0 live births
- Parous: ≥1 live birth

Responses coded as “refused,” “don’t know,” or missing were treated as missing and excluded from analysis.

RHQ131 (ever pregnant) was reviewed to understand reproductive history but was not used to define parity, as it includes pregnancies that did not result in live birth.

Menstrual Irregularity

Menstrual irregularity was defined based on self-reported questionnaire responses indicating irregular or inconsistent menstrual cycles, where consistent definitions were available across survey cycles.

Amenorrhea / Oligomenorrhea

Where available (e.g., RHQ074 and related variables):

- Amenorrhea: absence of menstruation for an extended period
- Oligomenorrhea: infrequent menstrual cycles

Definitions were harmonized across cycles where possible.

Subfertility

Subfertility was defined based on time to pregnancy:

- Participants reporting attempts to become pregnant for more than 12 months without success were classified as having subfertility

Participants with missing or inconsistent responses were excluded from this analysis.

Table 6* *Outcome definitions and coding*

Outcome	Definition	Coding
Parity	Number of live births (RHQ171)	0 = nulliparous; ≥1 = parous
Ever pregnant	Self-reported pregnancy history (RHQ131)	Yes/No (not used for parity definition)
Subfertility	Attempted pregnancy for >12 months without success	Yes/No
Menstrual irregularity	Self-reported irregular menstrual cycles	Yes/No
Amenorrhea	Absence of menstruation for extended period	Yes/No
Oligomenorrhea	Infrequent menstrual cycles	Yes/No

APPENDIX C. URINARY PHTHALATE METABOLITES AND DERIVED EXPOSURE MEASURES

This appendix provides details on exposure measurement and data processing.

Biomarker Measurement

Urinary phthalate metabolites were measured in NHANES using standardized laboratory methods. These biomarkers reflect recent exposure due to the short biologic half-lives of phthalates.

Log Transformation

Phthalate concentrations were right-skewed and therefore log-transformed prior to analysis:

- Natural logarithm (ln) transformation was applied
- This improves distributional properties and reduces the influence of extreme values

Creatinine Adjustment

Urinary creatinine (URXUCR) was included as a covariate in all regression models to adjust for urine dilution.

Cumulative Exposure Index

A cumulative phthalate exposure index was constructed for exploratory analysis by:

- Standardizing individual metabolite concentrations
- Summing standardized values across metabolites

This approach assumes equal contribution across metabolites and does not account for differences in biological potency or potential nonlinear interactions. Therefore, it is best interpreted as an exploratory composite exposure summary rather than a biologically weighted index.

APPENDIX D. SUPPLEMENTARY SENSITIVITY ANALYSES

This appendix summarizes additional analyses conducted to assess the robustness of the primary findings.

Alternative Exposure Specification

Models were re-estimated using alternative approaches, including:

- Creatinine-corrected exposure measures
- Comparison with models including creatinine as a covariate

Outlier Exclusion

Sensitivity analyses were conducted excluding extreme exposure values to assess the influence of outliers on model estimates.

Stratified Analyses

Stratified analyses (e.g., by age group) were explored to assess whether associations differed across subgroups.

Summary

Across sensitivity analyses, the association between MBzP and parity remained generally consistent in direction, although effect estimates varied slightly depending on model specification.

APPENDIX E. SAMPLE SIZE SUMMARY

Table 7* *Eligible and final analytic sample sizes across study analyses*

Analysis	Eligible Sample (N)	Final Analytic Sample (N)
Primary analysis (parity)	3,236	1,806
Subfertility analysis	2,424	2,275
Menstrual irregularity	425	340

Eligible and final analytic sample sizes are presented for the primary parity analysis and secondary analyses. Eligible samples include participants with available exposure data, while final analytic samples reflect complete-case inclusion based on outcome and covariate availability. The menstrual irregularity analysis was restricted to the 2013–2014 cycle due to data availability.

APPENDIX F. REPRODUCIBLE R CODE FOR DATA PROCESSING AND ANALYSIS

This appendix provides the primary R code used for data processing, variable construction, and statistical analysis in this study. The code is intended to support reproducibility and to provide a starting point for future researchers conducting NHANES-based analyses using survey-weighted methods. File paths and variable names may require modification depending on the user's local setup and NHANES data files.

The following R packages were used for data management and analysis:

```
library(tidyverse)
library(haven)
library(survey)
library(broom)
```

NHANES data were imported using the haven package. Multiple survey cycles were combined after harmonizing variable names and formats. The example below demonstrates data import and merging for a single cycle.

```
# Import NHANES datasets
demo_0304 <- read_xpt("DEMO_C.XPT")
phthalate_0304 <- read_xpt("L24PH_C.XPT")

# Merge datasets by participant identifier
nhanes_0304 <- demo_0304 %>%
  left_join(phthalate_0304, by = "SEQN")
```

The analytic sample was restricted to women aged 18–44 years who completed the Mobile Examination Center (MEC) examination.

```
nhanes_filtered <- nhanes_0304 %>%
  filter(RIDAGEYR >= 18,
         RIDAGEYR <= 44,
         RIAGENDR == 2,
         RIDSTATR == 2)
```

Parity was defined as a binary outcome based on the number of live births.

```
nhanes_filtered <- nhanes_filtered %>%
  mutate(parity_binary = ifelse(RHQ171 >= 1, 1, 0))
```

Urinary phthalate metabolites were log-transformed due to right-skewed distributions.

```
nhanes_filtered <- nhanes_filtered %>%
  mutate(
    log_MBzP  = log(URXMZP),
    log_MnBP  = log(URXMBP),
    log_MiBP  = log(URXMIB),
    log_MCPP  = log(URXMCP),
    log_MEHP  = log(URXMHP),
    log_MEHHP = log(URXMHH),
    log_MEOHP = log(URXMOH)
  )
```

NHANES subsample weights were adjusted for pooled cycles by dividing by the number of combined cycles (n = 6).

```
nhanes_filtered <- nhanes_filtered %>%
  mutate(phth_weight_pooled = WTSB2YR / 6)
```

Survey-weighted analyses were conducted using the survey package, accounting for clustering, stratification, and weights.

```
nhanes_design <- svydesign(
  id = ~SDMVPSU,
  strata = ~SDMVSTRA,
  weights = ~phth_weight_pooled,
  data = nhanes_filtered,
  nest = TRUE
)
```

Survey-weighted logistic regression models were used to estimate associations between phthalate metabolites and parity.

```
model_mbzp <- svyglm(
  parity_binary ~ log_MBzP + RIDAGEYR + BMXBMI + INDFMPIR +
    factor(DMDEDUC2) + factor(RIDRETH1) + URXUCR,
  design = nhanes_design,
  family = quasibinomial()
)
```

```
summary(model_mbzp)
```

False discovery rate (FDR) adjustment was performed using the Benjamini–Hochberg procedure as implemented in the base R function p.adjust().

```
primary_pvalues <- c(
  log_MnBP  = 0.633,
  log_MBzP  = 0.025,
  log_MCPP  = 0.687,
  log_MiBP  = 0.498,
  log_MEHP  = 0.226,
  log_MEHHP = 0.469,
  log_MEOHP = 0.338,
  log_sumDEHP = 0.170
)

fdr_adjusted_pvalues <- p.adjust(primary_pvalues, method = "BH")

fdr_results <- data.frame(
  exposure = names(primary_pvalues),
  raw_p_value = primary_pvalues,
  fdr_adjusted_p_value = fdr_adjusted_pvalues
)

print(fdr_results)
```

Sensitivity analyses were conducted to evaluate the robustness of the primary findings. The example below demonstrates exclusion of extreme exposure values.

```
nhanes_trimmed <- nhanes_filtered %>%
  filter(log_MBzP < quantile(log_MBzP, 0.99, na.rm = TRUE))

nhanes_design_trimmed <- update(nhanes_design, data = nhanes_trimmed)

model_trimmed <- svyglm(
  parity_binary ~ log_MBzP + RIDAGEYR + BMXBMI + INDFMPIR +
    factor(DMDEDUC2) + factor(RIDRETH1) + URXUCR,
  design = nhanes_design_trimmed,
  family = quasibinomial()
)

summary(model_trimmed)
```

The code presented here reflects the primary analytic workflow used in this study. Minor modifications may be required depending on NHANES cycle-specific file structures, variable naming conventions, and software versions. Users are encouraged to consult NHANES documentation for detailed guidance on data preparation and weighting procedures.

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VITA

Anusha Thondepu received a Bachelor of Pharmacy from KVSR Siddhartha College of Pharmaceutical Sciences, India. She completed her Master of Science degree at the University of Missouri–Kansas City in 2026.

During her graduate studies, she developed an interest in environmental epidemiology and population-based research. Her thesis focused on examining the association between urinary phthalate metabolites and reproductive outcomes among U.S. women using data from the National Health and Nutrition Examination Survey (NHANES). This work involved the application of survey-weighted statistical methods, data harmonization across multiple survey cycles, and interpretation of complex exposure–outcome relationships.

In addition to her academic training, she gained experience working with large public health datasets and developing reproducible analytical workflows. Her broader research interests include environmental exposures, women's reproductive health, and the use of data-driven approaches to better understand population health.